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New secure and efficient substitution and permutation method for audio encryption algorithm

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Abstract

A new substitution and permutation method for the encryption/decryption of two-channel audio files based on chaotic maps is presented. The proposed scheme's structure depends on encoding the original audio data into a new data range. Each value in the resulted range was translated to a binary sequence and both substitution and permutation operations are accomplished using the chaotic state and chaotic parameters. The substitution was attained using the XORing operation and Bernoulli substitution while the permutation method was based on the numbers sequence generated using a hyper chaotic system. A new key generation algorithm is proposed based on the properties of the square root of large prime numbers and a hyper chaotic system to produce the keys. The proposed cipher system was tested by various audio file sizes of (.WAV) extension. Performance evaluation and security measures to assess the algorithm's security against various attacks. The results of the key space analysis, statistical analysis, mean square error analyses, Peak signal to noise ratio (PSNR) analyses, entropy analyses, homogeneity, contrast, SNR, etc. showed that the proposed cipher is notably secured in opposition to the attackers and possesses an effective diffusion and confusion mechanism for better audio communication with inside the area of telecommunication.

Keywords Two-channel audio · Audio encryption · Audio decryption · Bernoulli map · Hyper chaotic system · A prime number

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1 Introduction

Researchers in audio encryption systems have been drawn to improving practical techniques to save and send data due to the rapid advancement of information technologies. To keep digital, particularly, reassured and secure audio data, it is important to change the format of the digital signals in a way that uncertified users cannot access real information. Keeping private information safe is a critical matter, and researchers are competing to develop algorithms that can ensure the protection of audio files in particular. Random keys are utilized as a part of security procedures for storing and transmitting digital data and they should have acceptable statistical characteristics and efficiency are required.

The difference between audio signals and images is huge. Audio signals have a large size of data samples with strong correlations. In addition to that, new audio or voice applications have real-time processing and need lower computation time and power consumption [1]. Permutation methods are insufficient to separate the correlation between adjacent samples in an audio stream.

Audio data is extremely important in multimedia data. The user's privacy will be jeopardized if it is not securely encrypted, which would have a significant impact. As a result, researching audio encryption schemes has become a popular academic subject. The main goal of the paper is to design a new audio cryptographic system for two channel audio signals. The audio data are encoded from the original data range to a new data range. Each number in the resulting range is converted to a binary sequence and both substitution and permutation operations are attained based on the Bernoulli map, hyper chaotic system. Because chaos is extremely sensitive to inherent conditions, it was used extensively in the development of the proposed cryptographic audio cipher.

1.1 Related works

Chaotic algorithms are more successful in most applications such as (generating pseudo-random numbers and performing multimedia encryption) because they have properties including high sensitivity to the initial values, no periodicity, pseudo-randomness, and ergodicity of chaotic sequences [2]. Recent methods for making the cipher confusion and diffusion procedures adjustable and unknown to attackers are described. One of these methods is proposed by Sheela et al. [3] proposes a novel audio cryptography algorithm based on chaotic maps, the hybrid chaotic shift transform (HCST), and the encoding principles of deoxyribonucleic acid (DNA). Chaotic maps, such as the 2D altered Henon map and the conventional map, are also used in the scheme. To execute HCST, the 2D altered Henon map is used, which owns an advanced chaotic manner for a wide variety of tuning options. The DNA encoding process is employed as an extra parameter to improve the cryptosystem's security. A new audio file encryption technique depending on a mix of a block cipher and chaotic maps is proposed in [4]. The blocks of data, everyone with a size of 512 bytes, are encrypted and decrypted by the suggested approach. Each block was fed into one

of three stages: permutation, Xor addition, and substitution. The permutation works on an audio block utilizing a chaotic Tent map in the permutation step. After that, the Xoring process works on the produced block, and the resulted block is finally substituted utilizing a multiplication inverse substitution algorithm. In [1], for real-time audio applications, a novel encryption approach was based on three chaotic maps. These chaotic maps were combined to increase resiliency where the work of the three steps including cyclic shift, permutation, and substitution of bits is replicated for four runs. The nonlinearity of confusion is obtained by the implementation of Mordell elliptic curves and some permutation processes. A Synchronization of a 4-D nonlinear fractional order hyper chaotic system with an external disturbance is a useful model for encryption and decryption algorithm that was used in [5]. The fractional order $0 < \alpha \leq 1$ is considered in this study to develop finite-time control for the synchronization of the sender and receiver systems. The audio signals are encrypted and decrypted using a random mask derived from a chaotic mask. The fractional order hyper chaotic system's path of error system is proven to be far superior to the classical one. To ensure information security, a novel audio encryption approach is presented [6]. The first step is to create a 1D not compressed audio file with a size of 16-bit of even length. Subsequently, the secret key is created by XORing the chaotic Henon map and the tent map to produce a pseudo-random number sequence. For generating a cipher audio file, the secret key is XORed with the original audio file. The study in [7] proposed a voice encryption method based on three separate schemes to boost the algorithm's flexibility and strength. To provide a secure and quick solution for voice encryption, the suggested approach employs an innovative encoding scheme, DNA encryption, and a permutation function. The security of any cipher system is based on the key.

Many key generation methods were proposed for generating the secret key for the cipher system based on different techniques. Several pseudo-random generators are being proposed in past years focused on chaotic maps [8–10]. The proposal in [11] developed a new PRNG using the Logistic map and the duffing map. The outputs of both maps utilized here are two-dimensional. They are based on combining several chaotic maps to increase their randomness. The suggested turbulence padding technique employs two chaotic maps, one of which is used to generate a random sequence and the other of which is used to dynamically turbulent the first map. In [12], turbulent padded chaotic map (TPCM) is a comprehensive framework for dealing with chaotic systems. In this study, they show how to use a variation in the output sequence to construct a simple and efficient messy system. They generate a new coding scheme for linear and nonlinear structures depending on producing a fresh key utilizing the Chebyshev 1D map and exponential map. The result, after that, was converted to machine words to meet the desired multicast range. The output of machine words thereafter is entered into the array, which was constructed to produce the key in the next step. A PRNG depending on a modified logistic chaotic system is suggested in [13]. This convergent chaotic system with fixed system parameters is analyzed and proven to exhibit chaotic behavior. To enlarge the key space and minimize the risk of an exhaustive attack, the parameter that is represented by floating point values, which are created by the system, is confused and reorganized to

increase the complication and randomness of altered PRNGs. The chaotic values generated by the preceding chaotic systems are related to the parameters of the following chaotic system, allowing the PRNG to produce sufficient results. By perverting the output succession and rearranging it, the system parameters of the former time cannot be obtained from the following time which guarantees security. A linear congruential pseudo-random number generator that uses chaotic maps and shift registers is present in [14]. LFSRs perform (XOR) sums on specific binary field positions at each stage. The purpose of this study is to generate a suitable pseudo-random sequence by using chaotic maps to randomly select positions at each stage and then performing a binary field summing operation on the least significant bits generated by a linear congruential generator. The paper [15] develops a new pseudo-random bit generator by incorporating chaos maps, a classic pseudo-generator, and a hash function to produce the random sequence. They use an exponential chaotic map, the RIPEMD-160, and linear and multiplicative congruential generators. In this generator, the exponential map was used to create messy numbers by using them as operators in linear and multiplicative congruential generators. These messy numbers are translated into binary sequences and used as initial inputs for RIPEMD-160 that results in a sequence of hexadecimal numbers. These numbers are permuted depending on a new shifting method. A new method to generate a random sequence is defined in [16] where a multi-level function is created by integrating Hénon chaotic map and the rational chaotic map. The resulting model creates two chaotic 64-bit binary strings and XORing them. Then, the resulting string is converted to 8 integer numbers in one round. The paper [17] provides a symmetric audio encryption method based on blocks (size 64 bytes). The complete technique is built up in three parts, with the first phase being the cyclic shifting of each block, the second phase being ciphering, and the third and final phase being the shuffling of the ciphered bytes. This reduces the adjacent correlation of plain audio. Piecewise linear chaotic map (PWLCM) and elementary cellular automata (ECA) are utilized to replace each block of shifted bytes, and PWLCM is also used in the final stage. The preceding key block and the preceding encrypted block are also used to derive each unique key block in addition to the secret key, which is thought of as a pair of the 64-byte external key and the plain audio-dependent value.

New multimedia applications and inexpensive instruments, according to our knowledge, require a real, secure, and efficient audio encryption technique. An efficient and safe candidate capable of giving high levels of protection with little complication and easy hardware accomplishment is required for modern applications like voice over internet protocol security (VoIP).

1.2 Contribution

When compared to existing audio encryption systems, the proposed methodology makes numerous advances toward achieving a high level of adequacy and security. These contributions are pointed out as follows:

- The proposed cipher provides high levels of protection with little complexity and easy hardware implementation that can be considered as an efficient and safe candidate for modern audio communication applications.
- The proposed algorithm is suitable for a 16-bit audio file where the audio data range is encoded into another range and then doubled its size by dividing each number into two 8-bit integers.
- The diffusion stage is split into two phases (XORing with key and Bernoulli substitution), and the confusion stage handles sorting permutation. As a result, these activities increase the proposed system's complexity while decreasing the time required for each encryption process. The security of any cipher system depends on the key generation algorithm. So, the hyperchaotic system and the square root of large prime numbers are used to design a new random key generator to create the random key sequence that is needed for diffusion and confusion stages.
- Furthermore, the suggested audio encryption approach has enough overall encryption performance and can be used to encrypt the ".wav" audio files of any size.

1.3 Organization

The organization of the article contains the following main subjects: section two introduces the basic theory of the paper. Then, section three describes the key generation algorithm of the special parameters and keys necessary for the encryption method. Section four describes in detail the proposed audio encryption method and decryption method. In Section five, the audio encryption method is tested below intensive performance and security analysis to demonstrate the hardiness and potency of this proposition. Finally, section six summarized the whole proposed work.

2 Basic theory

In this study two chaotic maps were utilized, a hyper chaotic system and a 1D Bernoulli map.

2.1 Hyper chaotic system

The new hyper chaotic system is created by combining the traditional Lorenz chaotic attractor system with a new variable, w, and a feedback controller $-\mathrm{d}x^3$. So, the hyper chaotic system is given as in the next relationship [18]:

$$\begin{aligned} \dot{x} &= a(y - x) \\ \dot{y} &= bx - y - xz + w \\ \dot{z} &= x^2 - cz \\ \dot{w} &= w - dx^3 \end{aligned} \tag{1}$$

where $\dot{x}, \dot{y}, \dot{z}, \dot{w}$ are the initial values and $a=24$, $b=25$, $c=3$, and $d=0.5$ are all constants greater than zero. There are two positive Lyapunov exponents ($\lambda_1=0$, $\lambda_2=2.5274$, $\lambda_3=2.1036$, $\lambda_4=16.3014$) over a wide range of parameters, implying that the system is hyper chaotic. Set the parameter of system as $a=24$, $b=25$, $c=3$, $d=0.5$, and initial values (1, 1, 1 ,1). As shown in Fig. 1, the system can represent the state of a hyper chaotic system [13].

2.2 1D Bernoulli map

The Bernoulli map is a 1D map that is expressed as follows [19]:

$$x_{k+1} = \begin{cases} rx_k + 0.5; x_k < 0 \\ rx_k - 0.5; x_k \geq 0 \end{cases} \tag{2}$$

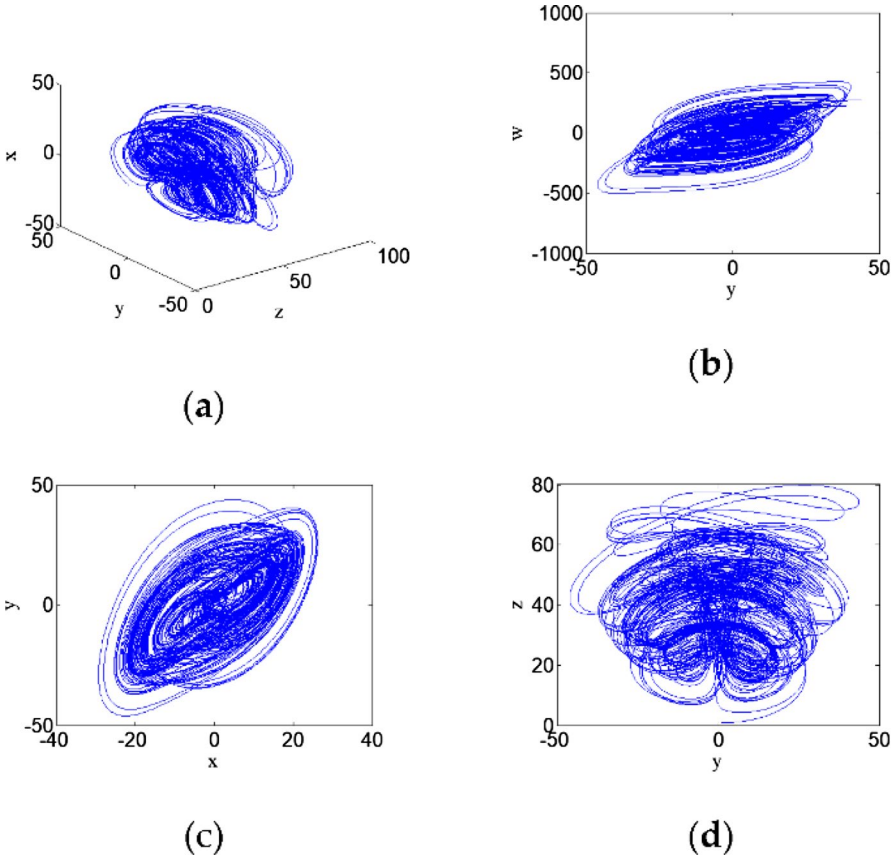


Fig. 1 **a** Hyper chaotic attractor; **b** hyper chaotic attractor on y–w plane; **c** hyper chaotic attractor on x–y plane; and **d** hyper chaotic attractor on y–z plane [13]

Here, $-0.5 < x < 0.5$ and $1.2 < r < 2$. A directed normalization method normalizes the Bernoulli map "x" from the range ($-0.5 < x < 0.5$) in Eq. (2) into the range ($0 < x \leq 255$) and $r = 1$. The map becomes [20]:

$$x_{k+1} = \begin{cases} r * x_k + 128 \,; x_k < 128 \\ r * x_k - 128 \,; x_k \geq 128 \end{cases} \tag{3}$$

The inverse of Eq. (3) is given by Eq. (4):

$$x_n = \begin{cases} \frac{x_{n+1}-128}{r} \,; x_{n+1} \geq 128 \\ \frac{x_{n+1}+128}{r} \,; x_{n+1} < 128 \end{cases} \tag{4}$$

3 Key generation algorithm

In the proposed audio cipher system, the random key sequence is created by a new key generation algorithm as shown in Fig. 2. The essential thought of the suggested key generator depends on taking advantage of the properties of the square root of large prime numbers for generating pseudorandom numbers. The key generation algorithm has the following steps:

- *Initial step* Four initial values (x_0, y_0, z_0, w_0) with four control parameter values are considered as the inputs to the hyper chaotic map Eq. (1). The initial values (x_0, y_0, z_0, w_0) are floating point numbers with a precision of 10^{-16}. The hyper chaotic map is iterated 100 times and the outputs are ignored so that the chaotic transitory effect is removed.
- *Prime number generation step* Repeat the following steps until eight prime numbers are generated:-
- Iterate the Hyper chaotic maps one time.
- In every iteration, convert the resulting four floating points $(x_{(i+1)}, y_{(i+1)}, z_{(i+1)}, w_{(i+1)})$ into four integers numbers (N_{v1};N_{v2};N_{v3};N_{v4}) using the following equations:

$$Fp = \text{ value } - \text{ floor}(\text{ value }) \tag{5}$$

$$Nv_i = \text{ Truncate } \left(Fp * 10^8\right) \tag{6}$$

where *value* is any one of the four floating points ($x_{(i+1)}$, $y_{(i+1)}$, $z_{(i+1)}$, $w_{(i+1)}$), Fp is the fraction part of the *value*, and N_{vi} represents one of the four resulted in integer numbers (N_{v1}, N_{v2}, N_{v3}; N_{v4}).

3. Each one of the resulting numbers (N_{v1}, N_{v2}, N_{v3}; N_{v4}) will be checked if it is prime or not. If the number is prime, it will be inserted into 1D array $P(1 \times 8)$ else it is ignored.

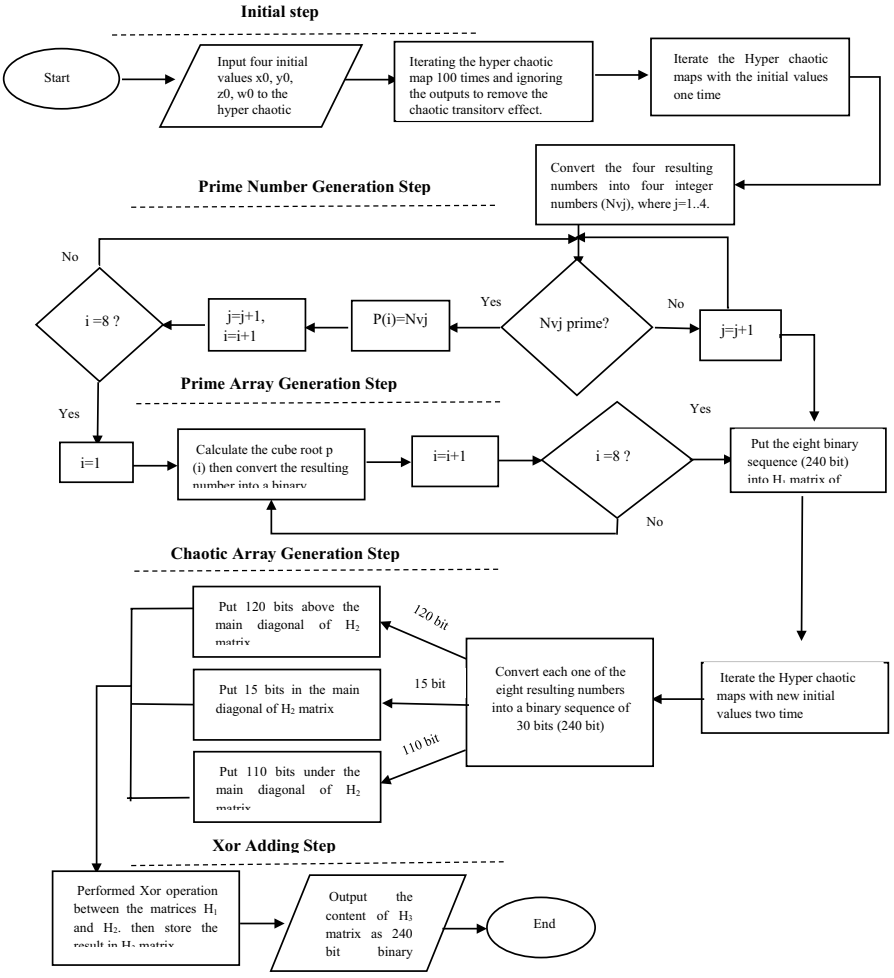


Fig. 2 Flowchart of one iteration one of the proposed key generation algorithm

- Prime array generation step:

1. For each number in the *P* array, do the following:

   a. Calculate the cube root and then convert the resulting number to a binary sequence of 30 bits in length.
   b. Put the resulting binary sequences in one-dimension array *B* (1 × 240).

2. Convert the result 1D B array into 2 D matrix *H*1 (15 × 16).

• *Chaotic array generation step* Iterate the Hyper chaotic maps Eq. (1) two times with four new initial values to generate eight random floating point outputs. Convert each output to a binary sequence of length 30 bits. The eight binary sequences are inserted into 2D array H_2 (15×16) using the following steps:

1. The first four binary sequences (120 bits) are put above the main diagonal of the H_2 matrix.x
2. Half of the fifth binary sequence (15 bits) is put in the main diagonal of the H_2 matrix.
3. The rest (15 bits) of the fifth binary sequence with the last three binary sequences (105 bits) are put under the main diagonal of the $H2$ matrix.

• *Xor adding step 5* Xor operation is performed between the H_1 matrix and H_2 matrix as shown:

$$H_3[i,j] = H_1[i,j] \oplus H_2[i,j] \tag{7}$$

where $i = 0,1,\ldots..15$ and $j = 0,1,\ldots..16$. H_3 is the resulting matrix of 240 bits in one iteration.

4 Performance analysis

The output series have to be secure to a great level. To determine the grade of security and examine the execution of the suggested key generator, some cryptographic tests must be performed such as the randomness test.

4.1 Randomness test

This investigation is carried out using the NIST statistical testing suite [21, 22, 23]. This testing is carried out using binary sequences generated from similar seed values. The tests were carried out by producing 1000 different binary sequences, each with a size of 10,000 bits. Various initial values are used to create each sequence. The NIST statistical software is used to test all 1000 sequences generated by the proposed key generator. The National Institute of Standards and Technology (NIST) has 15 examinations. A set of p-values is evaluated for each test and compared to a predetermined significance level of $\alpha = 0.01$ in each case (in other words, only 1% percent of the produced sequences are anticipated to be not passed). The ratio $\boldsymbol{\eta}$ is compared to a ratio that is considered acceptable where the range of acceptable parts is found by using the confidence interval specified as

$$\mathrm{p} \pm 3 \times \sqrt{\mathrm{p}(1-\mathrm{p})/\mathrm{n}} \tag{8}$$

where $p = 1 - \alpha = 0.99$ and n is the number of sequences. The NIST tests are accomplished on two kinds of sequences: individual sequences, and concatenate sequences.

The proposed key generator's randomness is investigated by dividing the NIST tests into two groups based on the length of generated sequences. The first group includes tests that must be conducted on sequences of length ≥ 100 bits, while the second group includes tests that must be done on sequences of length ≥ 100 bits. As a result, 1000 unique sequences with a total length of 1000 bits are constructed. The first test group is used to test the individual sequences. In addition, concatenate sequences are constructed, each of which is made up of 1000 separate sequences (i.e., the concatenate sequence has a length of 1000,000 bits). The first and second test groups are used to evaluate the concatenated sequences. For individual sequences, the ratio η is:

$$0.99 \pm 3 \times \sqrt{(0.99 \times 0.01/1000)} = 0.99 \pm 0.00944$$

That implies this ratio must be in the confidence range [0.98056–0.99944]. Tables 1 and 2 display the results of NIST tests performed on the two groups.

5 Proposed audio cipher algorithms

The main idea of the new cipher algorithm is based on doubling the audio data size by separating each audio two channel (16-bit) into two 8-bit numbers and performing the confusion and diffusion operations. This operation will decrease the correlation between the audio data and mark the algorithm as more complex. The audio cipher algorithm involves two main algorithms: audio encryption algorithm and audio decryption algorithm as shown in Fig. 3.

Table 1 The results of 1st Group of the NIST tests on the 1000 produced sequences for individual and concatenated sequences. the ratio η of *P*-value relates to individual sequences while the *P*-value relates to the concatenated sequences

Test name	η of Individual	Final result	*p*-value of Concatenate	Final result
Frequency	0.981	Pass	0.567067868785922	Pass
Block frequency	0.985	Pass	0.0194469137065036	Pass
Runs	0.991	Pass	0.769758236929016	Pass
Longest run	0.986	Pass	0.769758236929016	Pass
Rank	0.985	Pass	0.760719799799987	Pass
FFT	0.986	Pass	0.561657715013042	Pass
Non-overlapping	0.984	Pass	0.193322771289301	Pass
Serial (1)	1	Pass	0.571898782846506	Pass
Serial (2)	1	Pass	0.38614954603985	Pass
Cumulative sums	0.984	Pass	0.970523372339888	Pass

Table 2 The results of 2nd group of the NIST tests on the concatenated sequences.

Test name	*p*-value of concatenate	Final result
Overlapping	0.571898782846506	Pass
Universal	0.156051165200622	Pass
Linear complexity	0.9155644674782750	Pass
Approximate entropy	1	Pass
Random excursions (8 *p*-values)	0.4011591007749	Pass
	0.790484302579021	Pass
	0.788349096327006	Pass
	0.619086423740155	Pass
	0.529118931733297	Pass
	0.628085665031305	Pass
	0.796802642457138	Pass
	0.685117307856505	Pass
Random E-Variant (18 values)	0.358364780581687	Pass
	0.273863998914799	Pass
	0.285031757043786	Pass
	0.355311237687255	Pass
	0.303288852741251	Pass
	0.500897542457497	Pass
	0.832340750569455	Pass
	0.69621535124374	Pass
	0.299773104976262	Pass
	0.08261440854441483	Pass
	0.167729894509555	Pass
	0.64284495742406	Pass
	0.448284591852458	Pass
	0.266112690633499	Pass
	0.273844871520423	Pass
	0.26577379386964	Pass
	0.442324818656573	Pass
	0.603494628540826	Pass

The *P*-value relates to the concatenated sequences

5.1 Audio encryption algorithm

The proposed audio encryption algorithm involves three main steps: initialize step, encoding step, confusion step, diffusion step and decoding step (Table 3).

- Initialize step

  a. The input audio data is arranged as a one-dimensional array A with audio data length.
  b. The key generation algorithm is implemented with the initial values (x_0, y_0, z_0, w_0), which are floating point numbers wherein the precision is 10^{-16} and represented the cipher system keys.

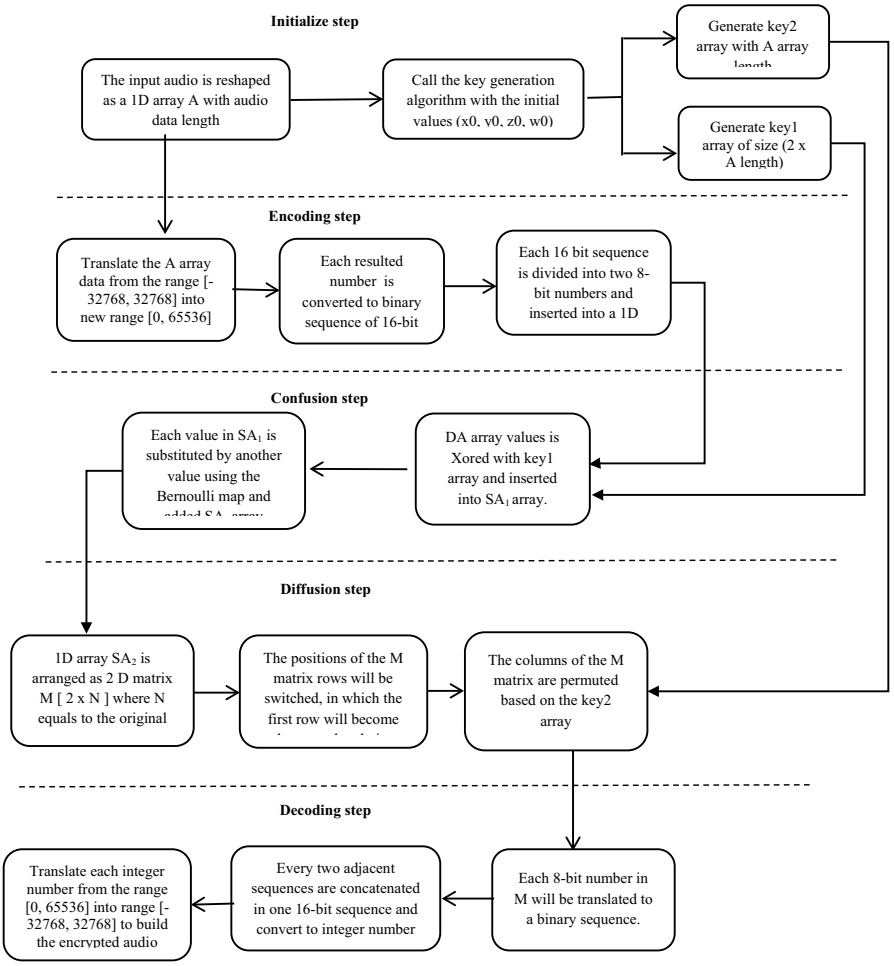


Fig. 3 Block diagram of proposed audio encryption algorithm

Table 3 Comparison of key space analyses with five other algorithms

	Proposed method	Ref. [3]	Ref. [4]	Ref. [1]	Ref. [5]	Ref. [6]	Ref. [16]	Ref. [25]
Keyspace	2^{240}	2^{336}	2^{319}	2^{266}	–	2^{249}	2^{512}	2^{128}

c. The generated keys are the key1 array of size ($2 \times$ audio length) that is needed in the confusion step, the key2 array with audio data length that is needed in the diffusion step.

- Encoding step

a. The audio data range [− 32768, 32768] is translated into another range [0, 65536] to make the encryption and decryption process easier. The translation method is performed using Eq. (9):

$$x_1 = x + 32768 \tag{9}$$

where x is a number its value is from -32768 to 32768 and its corresponding number (x_1) is from 0 to 65536.

For example: let the array $A[4]$ of original audio values is

− 280	− 276	− 313	− 314

Each value will be converted to the new range as shown:

32,488	324,929	32,455	32,454

Each resulting number ($x_{1)}$ is converted to a binary sequence of 16-bit. This sequence is divided into two 8-bit numbers and inserted into a one-dimensional array DA of size (2 × audio length).

b. In this step the resulting array length DA double of A array length which increases the complexity of the encryption process because the substitution and permutation operations will be done on the data that must have been fragmented and not the whole data. Return to our example, each encoded value will translate to a binary sequence of 16-bit binary as shown:

0,111,111,011,101,000	0,111,111,011,101,100	0,111,111,011,000,111	0,111,111,011,000,110

The 16-bit sequences after being divided into two 8-bit sequences will be:

01,111,110	11,101,000	01,111,110	11,101,100	01,111,110	11,000,111	01,111,110	11,000,110

The 8-bit sequences will be converted to integer values as shown:

126	232	126	236	126	199	126	198

• Confusion step

To increase the complication of the algorithm, the substitution process has been divided into two parts:

a. *Substitution by Xoring operation* Each value in the DA array is XORed directly with each value in the key1 array and inserted into the first substitution array SA_1.

Continue with our example, let the key1 array is:

17	127	70	125	14	89	58	25

Then, the result of performing the XOR operation between the 8-bit integer numbers array and the key1 array will be SA_1:

111	154	56	145	112	158	68	223

b. *Substitutions by using Bernoulli map* Each resulting value in array SA_1 is substituted by another value using the Bernoulli map Eq. (3) and added to the second substitution array SA_2.

Now, each value in the resulting SA_1 array will be substituted with another value using the Bernoulli map. For example, the first value in SA_1 array 111 which is <128 so the new value will be r * 111+128=239 where r =1 and so on for all SA_1 arrays. The result SA_2 array is:

239	23	184	17	240	30	196	95

• Diffusion step

In this step, the resulting one-dimensional array SA_2 from the previous step is arranged as 2 D matrix M in size 2 X N where N is equal to the original audio length. Then the matrix M is permuted in two steps:

a. *Permutated of rows* the matrix M consists of two rows, in this step, the positions of the rows will be switched, in other words, the first row will become the second and vice versa.

Continue with our example, let the M matrix is

239	23	240	196

The resulting M after the permutation of rows will be

23	17	30	95
239	184	240	196

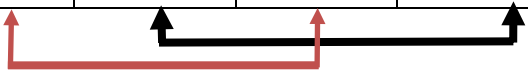
b. *Permutated of columns* the matrix M consists of N columns, in this step the columns are permuted based on the key2 array that is generated by the key generation algorithm. First of all, the key2 array is sorted in ascending order and the respective indexes are inserted into the index array. Then the columns are permuted based on the index array.

For our example, let the key2 array is

4	2	3	1

Then, every two columns will be exchanged, so the columns 4 and 2 then 3 and 1:

23	17	30	95
239	184	240	196



The result of permuted every two column array is:

30	95	23	17
240	196	239	184

• Decoding step
This step is the inverse of the encoding step where each value in the matrix *M* is a return to its original range:

a. Each 8-bit number resulting from the previous step will be translated to a binary sequence. Then, every two adjacent sequences are concatenated in one 16-bit sequence. For our example, the resulting array of 16-bit sequences is:

000,111,011,110,000	0,101,111,111,000,100	0,001,011,111,110,111	0,001,000,110,111,000

b. Convert each 16-bit sequence into an integer number and then return to its original range using the following equation:

$$x = x_1 - 32768 \tag{10}$$

Continue with our example, the result of converting every 16-bit sequence to an integer will be:

3824	24,516	6135	4536

Then, the resulting encrypted audio array after returning the original range is.

− 28,944	− 8252	− 26,633	− 28,232

5.2 Audio decryption algorithm

The decryption method is simply the opposite of the encryption method. The decryption method can be conveniently done using the pre-shared keys and can be

represented succinctly: initialize step, encoding step and decoding step are the same as in the encryption algorithm. The inverse diffusion step is the same in the encryption process except for the sequence of steps where the permutation of columns is performed before the permutation of rows. In the inverse confusion step, the substitution by XORing is performed before the substitution by Bernoulli which is used in the inverse Bernoulli map Eq. (4).

6 Results and discussion

The performance testing measurement for the proposed audio cipher has been implemented using the processor: Intel(R) Core(TM) i7-7500U CPU @ 2.70 GHz, RAM:8.00 GB (7.87 GB usable), system type:64-bit operating system, × 64-based processor and MATLAB R2018a. All the performance measurements are performed on five different2 channel WAVE audio files of different sizes. The WAVE file format is given as follows: (The WAVE file format, starts with a file header followed by a sequence of data chunks. A WAVE file has a single "WAVE" chunk which consists of two sub-chunks:

- An "fmt" chunk—specifies the data format
- a "data" chunk—contains the actual sample data ())

6.1 Brute force attack analysis

The key space of the suggested audio encryption algorithm is built from four initial values (x_0, y_0, z_0, w_0), which are floating point numbers with a precision of one decimal place (10^{-16}) [17]. A brute-force attack is unfeasible when the suggested approach has a vast key space. A key space of less than 2^{128} is considered insecure [24, 20]. For the suggested approach, the entire key space is $(10^{16})^4 \approx 2^{240}$. The encryption approach can withstand any brute-force attacks because the key space is large enough. The key space size of the submitted algorithm is compared with other algorithms in references [1–6, 16, 25].

6.2 Statistical attack analysis

To be sure that the proposed audio cipher has a strong immunity to statistical attacks, the tests such as histogram analyses, correlation analyses and entropy analyses are examined. These tests are examined for each step in the proposed algorithm and then compared with other references.

6.2.1 Histogram analysis

To observe the influence of encryption, the histograms of two audio files of varying sizes are plotted before and after each step of the encryption algorithm. Figure 4 shows the histogram of the original audio file, the encrypted audio file using only

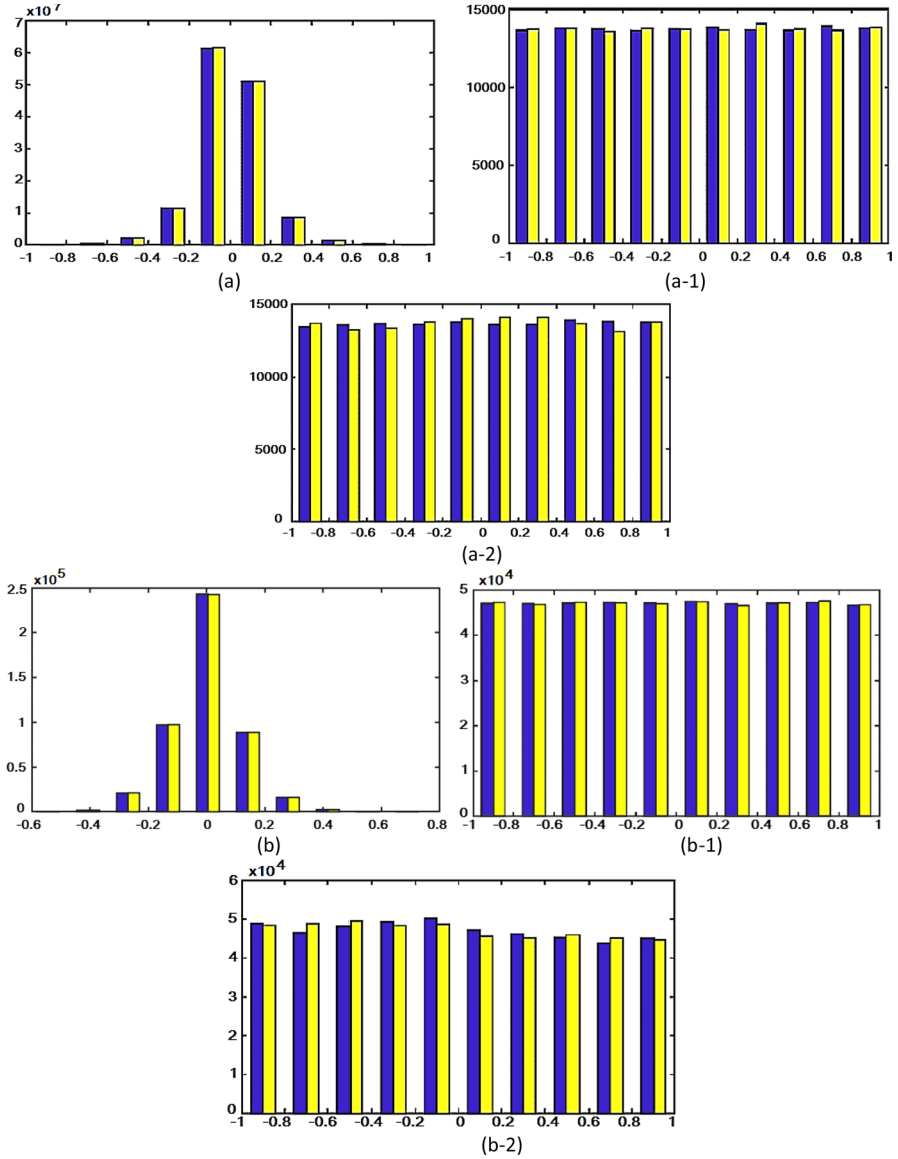


Fig. 4 Histogram analysis of **a** Original file1.wav. **a-1** Histogram of encrypted file1.wav after confusion step only. **a-2** histogram of encrypted file1.wav after confusion and diffusion steps. **b** Original file2.wav. **b-1** Histogram of encrypted file2.wav after confusion step only. **b-2** Histogram of encrypted file2.wav after confusion and diffusion steps

the confusion step and the encrypted audio file using the confusion and diffusion steps. From Fig. 4, the histograms of encrypted audio files (in two cases with confusion only or confusion and diffusion) are fully uniform and substantially different

from the original audio histograms, and so convey no statistical information about the original audio.

6.2.2 Correlation evaluation

The auto-correlation action locates the chaotic system, which generates secure encryption. The correlation coefficient between identical bits in the original and the encrypted signals is a helpful metric for evaluating the encryption quality of any cryptosystem. It is calculated as follows [27]:

$$r_{xk} = \frac{C(x,k)}{\sqrt{V(x)}\sqrt{V(k)}} \tag{11}$$

The covariance between the plain signal x and the encrypted signal k is $C(x, k)$. The variances of the signals x and k are $V(x)$ and $V(k)$. The variance $V(x)$ is calculated as follows:

$$V(x) = \frac{1}{N_2}\sum_{i=1}^{N_4}(x(i) - E(x))^2 \tag{12}$$

$$E(x) = \frac{1}{N_2}\sum_{i=1}^{N_d}(x(i) \tag{13}$$

$$C(x,k) = \frac{1}{N_x}\sum_{i=1}^{N_s}(x(i) - E(x))(k(i) - E(k)) \tag{14}$$

where Ns is the number of speech samples. The low value of the correlation coefficient rxk shows encryption with good quality. The correlation coefficients of five audio files of different sizes are calculated in four cases. In the first case, the correlation is calculated for the original audio file. In the second case, the correlation is calculated for the encrypted audio file with the confusion step only. In the third case, the correlation is calculated for the final encrypted audio file with confusion and diffusion steps. Lastly, the correlation is calculated between the original audio file and the final encrypted audio file. These cases are shown in Table 4, which illustrates the encryption at every step. It is clear that any step has low correlation results than the previous step. The correlation results for the encrypted audio files using confusion only are high while the correlation results for the encrypted audio file using confusion and diffusion approach to zero. The latter results prove that the strength of the given encryption system appeared clearly when performing the confusion and diffusion operations and the attacker cannot access any worthy data by creating a statistical attack.

Table 5 shows the comparison of correlation analyses of the proposed audio algorithm with the other seven other algorithms. The comparison results indicate that the correlation coefficient of the proposed system owns the lowest correlation

Table 4 The results of correlation analysis of the proposed audio encryption algorithm

Audio files	Audio file size	Correlation of original audio file	Correlation results for the encrypted audio file with confusion step only	Correlation results for the final encrypted audio file with confusion and diffusion steps	Correlation results between original audio and final encrypted audio file
File 1	535 KB	1	0.7466	0.0037	− 0.0021
File 2	1.79 MB	0.8862	0.6197	0.00008	0.00002
File 3	2.66 MB	0.8215	0.3006	0.0010	− 0.0011
File 4	1.02 MB	0.7757	0.3282	0.0020	0.00006
File 5	1.70 MB	0.6455	0.3536	− 0.0023	− 0.00004

Table 5 Comparison of correlation analyses with seven other algorithms

	Proposed method	Ref. [3]	Ref. [4]	Ref. [1]	Ref. [5]	Ref. [6]	Ref. [16]	Ref. [25]
Correlation coefficient result	0.00006	0.000038	− 0.0013	0.0031	0.0049	0.0128	− 0.030	0.0024

Table 6 The results of information entropy analysis for the encrypted audio files

Audio files	Entropy value of the encrypted file with confusion step only	Entropy value of the final encrypted file with confusion and diffusion steps
File 1	7.9991	7.9993
File 2	7.9987	7.9998
File 3	7.9988	7.9991
File 4	7.9994	7.9996
File 5	7.9997	7.9998

Table 7 Comparison of information entropy analysis with seven other algorithms

	Proposed method	Ref. [3]	Ref. [4]	Ref. [1]	Ref. [5]	Ref. [6]	Ref. [16]	Ref. [25]
Information entropy	7.9998	–	4.9819	7.9473	–	5.0155	–	–

value of all systems. This is proof of reducing the correlation between the components of the audio file and thus it became difficult for the attacker to hack the algorithm.

6.2.3 Information Entropy

The degree of uncertainty is determined using the information entropy technique. To avoid statistical attacks, a greater entropy value is desirable. The audio file that was processed by the encryption algorithm has an entropy magnitude given by Eq. (11) [6, 28]. In it, pv is the top magnitude of the audio data and $p(i)$ is the probability of value i.

$$\text{entropy} = -\sum_{i=1}^{\text{pv}} \left(p(i)\log_2(p(i))\right) \tag{15}$$

Table 6 shows the information entropy for five different encrypted audio files in four cases. From these results, it is observed that the entropy magnitude for the unprocessed audio files is increased after each encryption step. So, it is understood from this clear observation, the action of the given encryption system is in all stages. In turn, the presented algorithm will enhance the randomization of the values and reduce the chance of attacks.

Table 7 presents the comparison of entropy analyses of the suggested audio algorithm with the other seven other algorithms. The comparison results indicate that the entropy value of the proposed algorithm has a higher value than all algorithms and subsequently the system will improve the randomization of the values and reduce the chance of attacks.

6.3 Differential attack analysis

A successful encryption system must have the needed property of spreading the influence of a single plaintext bit over as a lot of the encrypted text as feasible, masking the plaintext's statistical texture. This means that even a minor variation in the original audio might produce a significant variation in the encrypted audio file. As a result, the differential attack is less effective and useless in practice. Mean square error (MSE), unified average changing intensity (UACI), and number of pixels change rate (NPCR) are the popular measurements that analyze the influence of a small change in the original audio data [29, 30].

1. *Mean square error (MSE)* MSE quantifies the avalanche effect, which occurs when the entire squared lag between any two cipher-audios is revealed. If X and Y are two vectors containing the two audio streams, the MSE between them is calculated by [31, 32, 33, 34]:

$$\mathrm{MSE} = \frac{1}{\mathrm{N}} \sum_{\mathrm{i}=0}^{\mathrm{N}-1} [X(\mathrm{i}) - \mathrm{Y}(\mathrm{i})]^2 \tag{16}$$

Here, N is the vector length of X (or Y). MSE value is commonly presented in dB. If the MSE value is equal to 30 dB, in turn, the quality variance of the two audios is high and the 1st audio is distinguished from the 2nd one. In this case, the algorithm owns an acceptable resistance to the differential attack.

2. *The number of pixels change rate (NPCR):* It is good when marginal changes in the original audio data produce a high change rate in the encrypted audio data. If the two encrypted audio vectors are X and Y, the original audios are only slightly different. Also, if D has a size equal to X or Y, so $D(i)$ is given by $X(i)$ and $Y(i)$ in Eq. (17).

$$\mathrm{NPCR} = \frac{\sum_{i,j} \mathrm{D}(\mathrm{i})}{\mathrm{W}} \times 100\% \tag{17}$$

where W represents the length of X or Y. It should be noted that Eq. (17) is at the condition of $X(\mathrm{i}) = \mathrm{Y}(\mathrm{i})$ then $\mathrm{D}(\mathrm{i}) = 0$ Else $\mathrm{D}(\mathrm{i}) = 1$ [26].

3. *Unified average changing intensity (UACI)* It calculates the mean contrast intensity between the unencrypted audio and encrypted one. If one calls the two

Table 8 Result of MSE, NPCR and UACI Analysis for the audio encrypted files

Audio files	MSE (encrypted file with confusion step only)	MSE (encrypted file with confusion and diffusion steps	NPCR (encrypted file with confusion step only)	NPCR (encrypted file with confusion and diffusion steps)	UACI (encrypted file with confusion step only)	UACI (encrypted file with confusion and diffusion steps)
File 1	39db	42db	98.360	99.604	16.02	33.33
File 2	39db	42db	98.354	99.604	16.04	33.33
File 3	39db	42db	98.372	99.602	16.03	33.21
File 4	39db	42db	98.345	99.597	16.02	33.39
File 5	39db	42db	98.361	99.617	16.03	33.40

Table 9 Comparison of MSE, NPCR and UACI with seven other algorithms

	Proposed method	Ref. [3]	Ref. [4]	Ref. [1]	Ref. [5]	Ref. [6]	Ref. [16]	Ref. [25]
MSE value	42	–	43.38	37.4943	$1.3146e+03$	–	–	–
NPCR value	99.617	–	–	99.6201	100	99.9995	–	100
UACI value	33.40	–	–	25.7984	–	–	–	33.3782

encrypted audio vectors X and Y, with just minor differences in the source audios, in this situation UACI is given by [35, 36]:

$$\mathrm{UACI} = \frac{1}{N}\left[\sum_{i,j}\frac{|\mathrm{X}(i)-\mathrm{Y}(i)|}{255}\right]\times 100\% \tag{18}$$

The typical NPCR magnitude is 100%, and the typical UACI magnitude is 33.3%.

Table 8 displays the outcomes of MSE, NPCR and UACI. The outcomes are so close to mentioned typical values that lead to the idea that the presented algorithm is solid against differential attacks.

Table 9 shows the comparison of MSE, NPCR and UACI analyses of the proposed audio algorithm with seven other algorithms. The comparison results in Table 9 indicate that the values MSE, NPCR and UACI of the proposed algorithm have higher values than all algorithms. This means this algorithm will improve random values and decrease the probability of attacks.

Table 10 The results of SNR analysis for the audio encrypted files

Audio files	SNR (original file)	SNR (encrypted file with confusion step only	SNR (encrypted file with confusion and diffusion steps)
File 1	− 6.8365	− 9.6638	− 34.2807
File 2	− 14.3481	− 16.1659	− 38.0568
File 3	− 15.1261	− 21.5395	− 21.7739
File 4	− 12.4475	− 17.7531	− 35.9567
File 5	− 15.3734	− 19.2705	− 37.5920

Table 11 Comparison of SNR Analysis with seven other algorithms

	Proposed method	Ref. [3]	Ref. [4]	Ref. [1]	Ref. [5]	Ref. [6]	Ref. [16]	Ref. [25]
SNR	− 38.2807	–	–	–	− 52.9039	–29.9476	− 50.49	–

Table 12 The results of PSNR analysis for the audio encrypted files

Audio files	PSNR value for the encrypted audio file with confusion step only	PSNR value for the encrypted audio file with confusion and diffusion steps
File 1	6.1005	7.3804
File 2	4.1601	5.1042
File 3	5.8534	6.6500
File 4	5.3088	6.4173
File 5	4.6266	5.9646

Table 13 Comparison of PSNR Analysis with seven other algorithms

	Proposed method	Ref. [3]	Ref. [4]	Ref. [1]	Ref. [5]	Ref. [6]	Ref. [16]	Ref. [25]
PSNR	5.1042	–	4.7170	10.6707	16.9428	4.2145	− 11.10	− 14.471

6.4 Encryption quality

The peak signal-to-noise ratio (PSNR), signal-to-noise ratio (SNR), and structural similarity index (SSIM) measurements are used to estimate the audio encryption framework's performance.

1. *Signal to noise ratio (SNR)* it specifies the signal quality, in addition to testing the performance of the encryption technique [37]. If the SNR value is low, the algorithm is stronger. The plain and encrypted audio data are required to perform this test. The SNR is in Eq. (19) depending on x_i and y_i to represent the matching sample values from audio files, and n is the number of samples [5, 6, 7]:

$$\mathrm{SNR} = 10\log_{10}\frac{\sum_{i=1}^{n} \times x_i^2}{\sum_{i=1}^{n}\left[x_i - y_i\right]}(d\mathrm{B}) \tag{19}$$

Table 10 shows that our suggested method produces low SNR values, indicating that it is a very powerful encryption method. Table 11 shows the comparison of SNR analyses of the proposed audio algorithm with seven other algorithms. The comparison of the outcomes refers to that the SNR value of the presented algorithm and reference [5] having lower values than other algorithms.

2. *Peak-signal-to-noise ratio (PSNR)* It is utilized to determine the audio quality. A good encryption strategy should result in a reduced PSNR value for the scrambled audio. PSNR is calculated mathematically using the following formula [1]:

$$\mathrm{PSNR} = 20\log_{10}\frac{255^2}{\sqrt{MSE}} \tag{20}$$

Table 14 The results of SSIM analysis for the audio encrypted files

Audio files	SSIM value or the encrypted audio file with confusion step only	SSIM value or final encrypted audio file with confusion and diffusion steps
File 1	0.0030	0.00001
File 2	0.0067	0.00005
File 3	0.0084	0.00005
File 4	0.0206	0.00125
File 5	0.03017	0.00182

Table 15 Comparison of SSIM analysis with seven other algorithms

	Proposed method	Ref. [3]	Ref. [4]	Ref. [1]	Ref. [5]	Ref. [6]	Ref. [16]	Ref. [25]
SSIM	0.00001	–	–	0.0002917	−2.8337e−08	–	–	–

Here, MSE is the average squared error of the plain and scrambled data. The outcomes of PSNR for the presented algorithm are given in Table 12.

Table 13 shows the comparison of PSNR analyses of the proposed audio algorithm with seven other algorithms. The comparison of outcomes refers to that the presented algorithm has a low value of PSNR which means that it has a good encryption quality.

3. *Structural similarity index (SSIM)* it means that the pixels have strong interdependencies, particularly when they are spatially closely separated. The mentioned conditions provide significant information about the organization of the visual scene's elements. The outcome is a decimal incentive that falls between − 1 and 1. The following expression determines SSIM mathematically [5]:

$$\mathrm{SSIM}(x,y)=\frac{\left(2\mu_x\mu_y+c_1\right)\left(2\sigma_{xy}+c_2\right)}{\left(\mu_x^2+\mu_y^2+c_1\right)\left(\sigma_x^2+\sigma_y^2+c_2\right)} \tag{21}$$

where μ_x is the mean value of x, μ_y is the mean value of y, ${\sigma_x}^2$ is the difference in x, ${\sigma_y}^2$ is the difference in y, σ_{xy} is the covariance of x and y, $c_1=(k_1L)^2$ and $c_2=(k_2L)^2$ are two variables to stabilize the division with weak denominator, L is the dynamic range of the pixel values, $k_1=0.01$ and $k_2=0.03$.

The outcome of SSIM measurements of the audio files for the presented algorithm is shown in Table 14.

Table 15 shows the comparison of the result of SSIM analyses of the proposed audio algorithm with other seven other algorithms. The comparison of the outcomes points to that the resulting SSIM value of the proposed algorithm has low value than all algorithms.

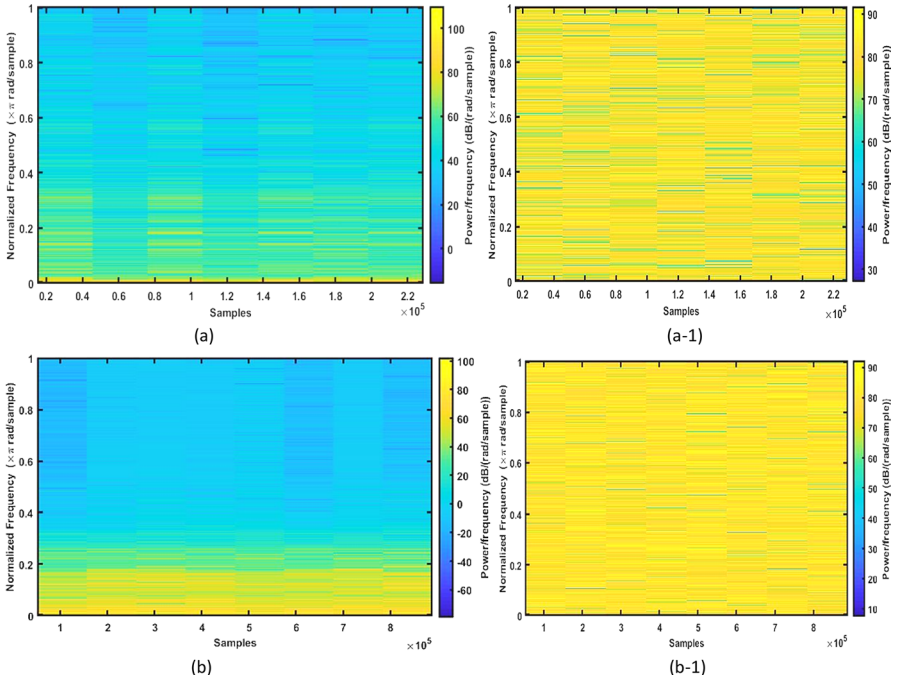


Fig. 5 **a** Spectrogram analysis of the 1st original audio file, **a-1** Spectrogram analysis of the encrypted 1st file, **b** Spectrogram analysis of the 2nd original audio, **b-1** spectrogram analysis of the encrypted 2nd file

6.5 Spectrogram of an audio signal

A spectrogram is a visual depiction of the frequency spectrum of an audio file that changes over time and is utilized to analyze audio signals. The audio signals are correctly encrypted if the spectrogram of the cipher audio is uniform [6]. Figure 5 shows a uniform spectrogram of the encrypted audio file in our proposed technique. This indicates that the original audio signals have been encrypted effectively.

Table 16 The results of contrast analysis for the encrypted audio files

Audio files	Contrast value (original file)	Contrast value (encrypted file with confusion step only)	Contrast value (encrypted file with confusion and diffusion steps)
File 1	0.1029	3.6463	10.4896
File 2	0.0827	3.4792	10.4682
File 3	0.1187	4.1284	10.3693
File 4	0.1580	4.3399	10.4651
File 5	0.1501	3.9133	10.4910

6.6 Contrast

Dissimilarity in audio signals is determined by the contrast level. It is a statistic for determining whether encryption is effective and whether it has the desired diffusion [1]:

$$\text{Contrast} = \sum_{i,j} |i-j|^2 P(i,j), \tag{22}$$

The position of a specific element of audio data is represented by i and j. $P(i,j)$ refers to the order of the co-occurrence matrix. The result of contrast values of audio data for the proposed algorithm as shown in Table 16 indicates that our proposal provides a good diffusion property.

6.7 Homogeneity

Depending on the type of audio file, the audio data is usually distributed. Homogeneity analysis is used to quantify the spread of components in a gray level co-occurrence matrix (GLCM) to determine distribution features. It can be stated mathematically as [1]:

$$\text{Homogeneity} = \sum_{i,j} \frac{P(i,j)}{1+|i-j|'} \tag{23}$$

Here, i, j and $P(i,j)$ are defined previously. Table 17 shows the homogeneity values of two different audio files, the plain audio file and the encrypted audio file. From these results, one can detect the high performance of the proposed algorithm.

7 Conclusion

Based on the chaotic property and cryptography properties confusion and diffusion, a new approach for audio cipher has been developed. From the results of the NIST statistical testing suit, it was observed that the hyperchaotic map has a highly chaotic and dissipative system, and 1D Bernoulli Map is an invertible

Table 17 Results of homogeneity analysis for the encrypted audio files

Audio files	Homogeneity value (original file)	Homogeneity value (encrypted file with confusion step only)	Homogeneity value (encrypted file with confusion and diffusion steps)
File 1	0.9491	0.5290	0.3905
File 2	0.9587	0.5317	0.3898
File 3	0.9410	0.5185	0.3919
File 4	0.9212	0.5140	0.3893
File 5	0.9250	0.5231	0.3898

equation, so one can use them in confusion and diffusion operations. The essential idea of the proposed approach is to decrease the correlation between two-channel audio files by separating the internal compound of the audio file. First of all, the audio data is encoded by converting its original range into another range. Then each value audio file is encrypted in main two steps (confusion and diffusion steps). A series of experiments were set up to assess the reliability of the proposed algorithm, including brute force attack, differential attack analysis, histogram, Spectrogram, information entropy analysis, and correlation analysis. Signal-to-noise ratio (SNR), peak signal-to-noise ratio (PSNR), mean square error (MSE), structural similarity Index (SSIM), signal-to-noise ratio (SNR), homogeneity, contrast. These tests are examined for the original audio file, the encrypted audio file with confusion only and lastly for an encrypted audio file with confusion and diffusion steps. The following are the outcomes of the analytical study of the suggested algorithm:

- The proposed system has a large key space which equals 2^{213} considering a large resistance to the Brute Force Attack. In addition, it can resist differential attacks.
- The encrypted audio file has a uniform histogram and uniform spectrogram.
- The performance of the proposed system is estimated through SNR test, and PSNR testing methods
- Low correlation coefficient value for encrypted audio files close to the ideal value of 0.
- Information entropies for the audio files have high values which means that no information available for the attacker.
- The proposed system has a good diffusion property which is tested through contrast level analyses.
- The coding and decoding steps in the encryption algorithm added a clear complexity to the encrypted audio files because they converted the audio data from one range to another range.
- More dimensions of a hyper chaotic system produce a larger keyspace, so more attack resistance.
- Exchanging the Bernoulli map with a nonlinear s box will increase the security of the encryption algorithm.

Finally, the observed security results demonstrate that the suggested system is efficient and capable of providing a grand grade of protection when it competes with other current audio encryption algorithms having a static or dynamic structure. As a result, the suggested cipher is a viable cryptographic alternative since it achieves a good balance between system performance and security. For future work, the proposed algorithm can be utilized in video encryption and decryption.

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Declarations

Competing interests Not applicable.

Ethical approval Not applicable.

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